

1)

- a) The pmos switch in Figure 1 is located between a purely resistive load and a battery. The resistance of the load is $10\ \Omega$ and the battery voltage is 48 V .

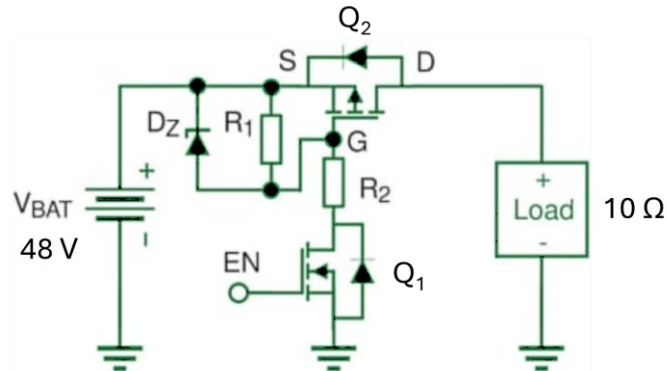


Figure 1. pmos switch between the battery and the load

R_1 is $5\text{ k}\Omega$ and R_2 is $19\text{ k}\Omega$. If a voltage (EN) is applied to the gate of nmos transistor Q_1 , which is much larger than the gate-source threshold voltage of Q_1 , Q_1 conducts and the $R_{ds(on)}$ of Q_1 can be neglected. The $R_{ds(on)}$ of the pmos Q_2 can be observed in Figure 2.

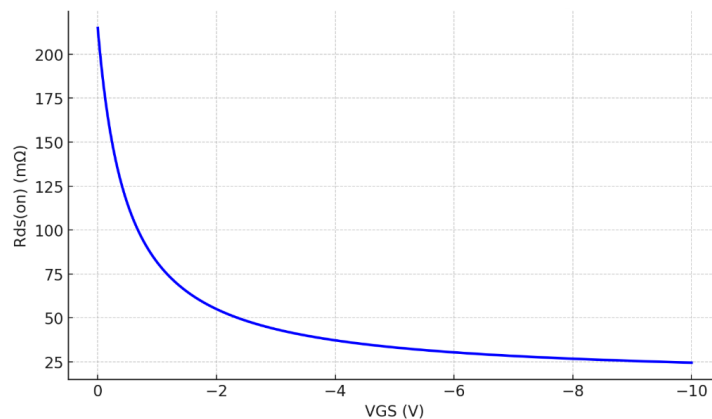


Figure 2. $R_{ds(on)}$ of pmos vs. V_{GS}

Calculate the current flowing from the battery to the load. What is the power dissipation of Q_2 ? $V_{Z,diode} = 10\text{ V}$.

- b) Q_2 is an SMD transistor and mounted on a PCB. If the thermal resistance between the junction of Q_2 and the heat sink is $2\text{ }^\circ\text{C/W}$ and temperature of the heat sink is assumed to be constant at $50\text{ }^\circ\text{C}$, what is the junction temperature of Q_2 ?

If the maximum rated temperature of Q_2 is $150\text{ }^\circ\text{C}$, what is the maximum rated power dissipation of Q_2 in order not to exceed the maximum rated junction temperature of Q_2 ? If the $R_{ds(on)}$ is $20\text{ m}\Omega$ for the maximum rated power dissipation of Q_2 , calculate the load resistor value.

- c) If the voltage applied to the gate-source of Q_1 is 0 V (Q_1 turned-off), which is much lower than the gate-source threshold voltage, calculate the power dissipation of the Q_2 . Use the $R_{ds(on)}$ values in Figure 2.

2)

- a) The linear regulator (LDO) in Figure 3 has an input supply voltage V_{IN} of 12 V. Determine V_{REF} and the resistor values R_1 and R_2 , such that the LDO delivers an output voltage of 5 V.

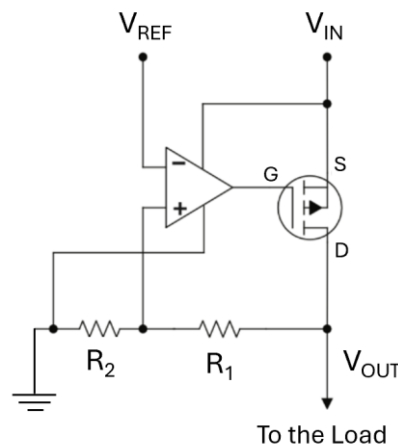


Figure 3. LDO with an operational amplifier and a pmos transistor

- b) Calculate the power dissipation of the pmos transistor for V_{IN} voltages of 8, 12 and 16 V and a load resistor of 10 Ω .
- c) Suggest a simple method to generate a V_{REF} voltage of 5 V.

3)

- a) Design a Buck Converter, which converts an input voltage of 15 V to 5 V. The Buck Converter shall incorporate a half-bridge with high-side and low-side MOSFETs. You do not need to implement the closed-loop configuration to adjust the output voltage.

Draw the PWM signal for high-side and low-side switches such that no shoot-through occurs. The switching frequency shall be 100 kHz. You do not need to calculate the output voltage ripple.

- b) Draw the currents flowing through the inductor and the high-side and low-side MOSFETs in steady-state if the inductor value is 50 μ H. The load resistor is 10 Ω .